

CENG381 Stochastic Processes

Biased Random Walk - Question Set 1

Notation used in this question set:

p	probability of stepping RIGHT (winning one unit) at each step
q = 1 - p	probability of stepping LEFT (losing one unit) at each step
N	right boundary (the 'winning' absorbing state)
R(k)	probability of reaching N before 0, starting from position k (where 0 is the 'ruin' absorbing state)
F(k)	expected number of steps until absorption (at 0 or N), starting from position k

Key formula (biased case, $p \neq q$): $R(k) = ((q/p)^k - 1) / ((q/p)^N - 1)$. When $p = q = 1/2$ (unbiased): $R(k) = k / N$ and $F(k) = k*(N-k)$.

Q1 (30 points). A gambler plays a biased coin-flip game with $p = 0.4$ (prob of winning each flip) and $q = 0.6$ (prob of losing). The game ends when the gambler's fortune reaches 0 (ruin) or $N = 5$ (win). The gambler starts at $k = 2$.

- Write the recurrence relation for $R(k)$, the probability of winning (reaching $N=5$) from position k , and state the boundary conditions $R(0)$ and $R(5)$.
- The characteristic equation for $R(k) = p*R(k+1) + q*R(k-1)$ has roots $r_1 = 1$ and $r_2 = q/p$. Compute q/p and write the general solution $R(k) = A + B*(q/p)^k$.
- Apply the boundary conditions to find A and B , then compute $R(2)$. Is $R(2)$ greater or less than $2/5 = 0.4$ (the fair-game value)? Explain the direction of the difference.

Q2 (35 points). Consider the same biased game ($p = 0.4$, $q = 0.6$, $N = 5$) but now ask: starting from $k = 2$, what is the expected number of steps $F(2)$ until the game ends?

The recurrence for $F(k)$ is non-homogeneous:

$$F(k) = 1 + p*F(k+1) + q*F(k-1) \quad \text{for } k = 1, 2, \dots, N-1$$

$$F(0) = F(N) = 0 \quad (\text{game already over at boundaries})$$

The homogeneous part has the same roots as the $R(k)$ recurrence.

A particular solution has the form $F_{\text{p}}(k) = c*k$ (try this form).

- Rearrange $F(k) = 1 + p*F(k+1) + q*F(k-1)$ into the standard form $p*F(k+1) - F(k) + q*F(k-1) = -1$. Substitute $F_{\text{p}}(k) = c*k$ into this equation and solve for c . (Recall $p + q = 1$, so simplify carefully.)
- Write the general solution $F(k) = A + B*(q/p)^k + c*k$, where A and B are free constants and c is the particular solution coefficient from part (a). Apply $F(0) = 0$ and $F(N) = F(5) = 0$ to find A and B .

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- c) Compute $F(2)$ using your formula. How does the expected game duration compare to the unbiased case $F_{\text{unbiased}}(2) = k(N-k) = 2 \cdot 3 = 6$?
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Q3 (35 points). The Fibonacci sequence $F(0)=0$, $F(1)=1$, $F(2)=1$, $F(3)=2$, $F(4)=3$, $F(5)=5$, ... is defined by the recurrence $F(n) = F(n-1) + F(n-2)$ for $n \geq 2$.

- a) Solve this recurrence using the characteristic equation method. The characteristic equation is $r^2 - r - 1 = 0$. Find both roots $r_1 = (1 + \sqrt{5})/2$ and $r_2 = (1 - \sqrt{5})/2$, and write the general solution $F(n) = A \cdot r_1^n + B \cdot r_2^n$.

- b) Apply the initial conditions $F(0) = 0$ and $F(1) = 1$ to determine A and B . This gives Binet's formula:

$$F(n) = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$$

- c) Use Binet's formula to show that $F(n+1)/F(n) \rightarrow (1+\sqrt{5})/2$ as $n \rightarrow \infty$ (the 'golden ratio' $\phi \sim 1.618$). Explain why the term involving $r_2 = (1-\sqrt{5})/2 \sim -0.618$ becomes negligible for large n .